

Innovation Brief: OFI Case Study

Predictive Return Management System for NexGen Logistics

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Link of the project:

<https://github.com/mdphaye/Predictive-Return-Management>

Overview

NexGen Logistics handles hundreds of deliveries daily across India, facing growing challenges in return management and cost control. While most logistics analytics focus on delivery optimization, very few systems proactively predict **which deliveries are most likely to be returned before they happen**. This project/case study introduces a data-driven solution that forecasts return probability and helps logistics teams take preventive actions in advance.

Problem Statement

- **Option 8:** Your Own Idea: Identify a unique problem from the data and create an innovative solution.
- Reverse logistics: the process of handling product returns, often leads to hidden operational costs, delayed asset cycles, and customer dissatisfaction. Currently, NexGen operates reactively, addressing returns only after they occur. This results in increased handling costs, wastage of packaging materials, and reduced delivery efficiency. **The company lacks a system to proactively identify high-risk orders before shipment.**

Proposed Solution

I propose a Predictive Return Management System (PRMS): a lightweight Machine Learning–powered web tool that predicts the likelihood of an order being returned using past order, customer, and delivery data. The system provides a Return Risk Score (Low, Medium, High) for every order and suggests corresponding preventive measures such as enhanced packaging, pre-delivery confirmations, or quality checks. This enables NexGen to move from reactive return handling to proactive prevention, improving both cost efficiency and customer satisfaction.

Technical Approach/Methodology

The model uses lightweight ML algorithms (Logistic Regression / Random Forest) trained on features such as order value, delivery time, product category, warehouse, and customer history. Key steps include:

- Data cleaning and feature engineering using Python (Pandas, NumPy)
- Model training and evaluation using scikit-learn
- Deployment as an interactive dashboard using Streamlit
- Real-time return risk visualization through tables and charts

The system is lightweight, easily deployable, and runs efficiently on standard hardware.

Expected Business Impact

- Reduce return-related costs by 10–15% through preventive actions
- Increase customer satisfaction via fewer return experiences
- Optimize packaging and logistics resource allocation
- Enable data-driven decision-making across departments
- Build NexGen’s reputation as an innovation leader in logistics

Key Features

- Return Risk Scoring: Predicts likelihood of return per order
- Interactive Streamlit Dashboard: Visualizes risks and recommendations
- Feature Importance Insights: Identifies main drivers of returns
- Scenario Simulation: Allows ‘what-if’ cost-saving analysis
- Lightweight Model: Quick training, low compute requirement

Future Enhancements

Future versions can integrate external data sources such as weather, product reviews, and regional demand fluctuations for more robust predictions. The system can also connect with the warehouse management system (WMS) to automate packaging adjustments for high-risk orders.

Tools & Technologies Used

- **Programming languages:** Python
- **Libraries and frameworks:** Pandas, NumPy, Scikit-learn, Streamlit, Matplotlib, Plotly, Joblib, Google Colab
- **Version control:** Github

Conclusions

The deployed Streamlit web app is accessible publicly. The tool allows operations/Logistics teams to upload new order batches in CSV format, get real-time 'Return Risk' predictions, visualize results and download outcomes for further action.